The Embedding ι — Working Document and Forcing Protocol

HLV time sector $\hookrightarrow$ FFGFT recursion (Doc. 299)

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Doc. 299 The Embedding ι – Working Document and Forcing Protocol (HLV time sector $\hookrightarrow$ FFGFT recursion)

Purpose and scope

Doc. 297 records the embedding

$$\iota : \text{ HLV base object } \hookrightarrow L^2(T^4) \otimes \mathbb{C}^3 \tag{.1}$$

as an open conjecture: structurally plausible, but not constructed. This document sets up the next clean step (P35) as a *protocol* – it fixes, *before* the joint work, the domain, the codomain, the candidate map, the matched null models, and the decision criterion. It is not a proof and not a claim about an outcome; it names what would have to count as *forced* and what as merely *allowed*, so that the later evaluation cannot be adjusted after the fact.

The frame is the *time sector*. Following the sharpening in the correction register (Doc. 190, R47), two senses of containment must be kept apart: *dimensionally*, HLV counts larger ($T^7 = T^4 \times T^3$, $7 = 4+3$; R43); *in content, in the time sector*, HLV $\subset$ FFGFT, because the HLV spiral-time window is a bounded section of the unrolled recursion time. In unified notation – both $D6 +$ time, HLV as $D6 +$ spiral-time, FFGFT in Hilbert space as $D6 +$ recursion-time – the window is not additive but internal. ι is exactly the map that makes this internal section explicit.

That it is a section takes no value from the window (Doc. 298): a sub-model that evaluates correctly on its scale is not refuted but located. ι returns to the section the compact whole of which it is the reduction.

Domain: the HLV side

Doc. 297 (§"Two HLV spiral-times") separates two distinct objects, both of which HLV calls "spiral-time". They are separate embedding candidates and must not be conflated in the protocol.

Object A – the geometric time. The A_5 singlet on the S^1 factor of the factorisation $T^7 = T^4 \times T^3$ (Doc. 285): decoupled, separable, without a memory channel. Purely geometric.

Object B – the temporal-dynamic spiral-time. The triadic operator $\Psi(t) = (U_1, U_2, U_3)$ with memory channel U_3 (Krueger, Consciousness document, Eq. 7): a bounded window accumulator with a fixed, sliding window M_m; older values actively drop out. In every representation – linear, polar, spiral – the dependency structure is the same: finite short-term memory with fixed horizon M_m, without accumulation (Doc. 297: the spiral representation is a coordinate choice, not a structural property).

Codomain: the FFGFT side

The codomain is the compact recursion in the Hilbert space $H = L^2(T^4) \otimes \mathbb{C}^3$ (Doc. 230/282, lossless bijection). Three structural elements are decisive for ι:

- **The memory kernel.** The Fourier transform of the discrete spectral density $\sum_k g_k^2\,\delta(\omega-\omega_k)$ (Doc. 283) – the full, accumulating memory structure of the recursion.
- **The frozen winding.** In the frozen configuration (Doc. 295) the winding closes after exactly $1/(100\,\xi_0) = 75$ turns; this is the smallest coherent unit of the recursion.
- **The K_{frak} recursion** $\xi_{n+1} = \xi_n(1-100\,\xi_n)$: the present state carries the entire history as a structural imprint; it generates the log-spiral with ratio e (Doc. 295). Memory as generative structure, not as storage.

The candidate map ι (two branches)

Matching the two objects of the domain, ι has two branches with different natural locations.

ι_{geo} **(Object A).** The A_5 singlet (S^1, decoupled) maps to the $\xi \to 0$ fixed point of the recursion – the geometrically separable, most information-poor configuration of the compact time (constant pitch, winding closes after 75). It is Archimedean from geometric factorisation, but carries no ratio e (Doc. 295 §10: blocked on the e-criterion).

ι_{dyn} **(Object B).** The window U_3 maps to the hard-reset kernel with finite support below one turn:

$$\iota_{\text{dyn}} : U_3 \;\hookrightarrow\; \left\{ K \in L^2(T^4)\otimes\mathbb{C}^3 \;\middle|\; K(t-\tau) = 0 \text{ for } t-\tau > M_m\cdot\Delta t,\; M_m < 75 \right\}. \qquad (.2)$$

This corresponds to the Markov limit of the memory kernel ($g_k \to 0$, Doc. 283), sharpened to a bounded window. Both branches are bounded sections (windows) of the unrolled recursion time; this is exactly the content-level HLV $\subset$ FFGFT of the time sector.

The information hierarchy (Doc. 297) locates the branches:

$$U_3 \;\subset\; \text{Markov limit} \;\subset\; \text{Archimedean} \;\subset\; \text{log-spiral}\,(e) \;=\; \text{FFGFT core}. \qquad (.3)$$

The geometric singlet lies to the side of this chain (Archimedean from factorisation, not from accumulation). Each step is more information-poor than the next; U_3 is the poorest.

The forcing question: forced or merely allowed

The core of the protocol. That a section *exists* into which U_3 embeds is weak – a bounded window operator can be placed into the degenerate boundary case of any sufficiently rich recursion. The substantive question is whether the embedding is *forced* or only *allowed*. This is the same forcing logic as in the C3A5 specificity test (Doc. 293/294), transferred to the time axis.

Forced (HLV $\subset$ FFGFT **in the time sector, genuine).** The Markov-limit section of the recursion *necessarily* coincides with the hard-reset kernel of U_3, and an independent invariant of the section agrees with U_3 where generic bounded window operators do not. ι is then the unique structural placement.

Allowed (mere compatibility). ι merely embeds compatibly: every bounded window operator would sit equally in the degenerate boundary case, and the match carries no content beyond the definition (P35).

Circularity caveat (P35). U_3 is by construction a hard-reset window. Reading it as the Markov-limit section of the recursion is therefore initially *definitional*, not an independent confirmation – just as the icosahedral object gives icosahedral reference distance zero because it *is* that object (Doc. 294). The forcing must therefore come from an *independent* signature that U_3 shares with the recursion section without both being defined by the same construction rule.

Matched nulls (time axis, ι_{dyn})

The forcing is tested against a family of matched null models – the temporal counterpart to axis shuffle and generic tight-frames (Doc. 293/294). It also mirrors Marcel's own OP8 null family (Doc. 297), so that the instruments interlock:

- **memory-free Markov** – the baseline that ι_{dyn} predicts in the boundary case;
- **phase-shuffled** and **time-shuffled** – destroy temporal structure while preserving marginal statistics;
- **coherence-only** and **information-only** – isolate single channels;
- **generic window operator** – random M_m, random kernel with the same support: the direct analogue of "random rotation axis / generic tight-frame";
- **generic ML baseline**.

Calibration as in the specificity test: an exact recursion section gives distance zero, generic window operators lie in a broad band. ι_{dyn} is *forced* exactly when U_3 sits at the zero point where the generic nulls lie in the band; *allowed* when every window null reaches the same distance.

Matched nulls (geometric branch, ι_{geo})

For the geometric branch the null family is the spatial counterpart – axis shuffle and generic frames in the sense of Doc. 293/294, here in the form agreed with the HLV side:

- **random C_3/A_5 embeddings** – the native carrier against arbitrarily rotated embeddings of the three-/five-fold structure;
- **alternative projection stars** – cut stars other than the icosahedral one;
- **tetra-/octahedron controls** – crystallographic point groups without five-fold symmetry as a negative control;
- **shuffled C_3 bases** – permuted fiber bases with preserved marginal statistics;
- **spectrally matched null carriers** – carriers with the same power spectrum but without the icosahedral axis structure.

Calibration as in the $C3A5$ specificity test (Doc. 294): the exact icosahedral projector gives distance zero to the reference spectrum (all fifteen angles 63.435°, redistribution $2/9$), generic frames lie in a broad band (median RMS ≈ 22). ι_{geo} is *forced* exactly when the native carrier sits at the zero point where these nulls lie in the band; *allowed* when generic embeddings reach the same distance.

Jointly booked architecture. Both null families are agreed with the HLV side. Krueger (IOTA-BRIDGE1 v0.1) tests the same finite proxy question – a native-6D spiral-time-window carrier as a C_3 fiber-compatible object in a finite $L^2(T^4) \otimes \mathbb{C}^3$ proxy – under spectral, C_3/A_5, U_3 time-window, and screen/information diagnostics. The claim boundary is tight on both sides: PASS = finite-proxy ι-forcing candidate (no subset proof, no physics/mass/QFT/GR derivation); BLOCK = this concrete finite embedding route is narrowed or blocked.

Decision criterion and falsification

The pre-registered OP8-Neuro test (Doc. 297) is the outer decision test; the forcing protocol adds the specificity step:

1. M_3 **does not beat the memory-free Markov baseline.** U_3 lies entirely in the degenerate boundary case. ι_{dyn} holds in the Markov limit – but from this alone follows *allowed*, not *forced* (every hard-reset window achieves this). The specificity run is additionally required for forcing.
2. M_3 **clearly beats the memory-free Markov baseline.** U_3 carries non-Markov structure (revivals, BLP backflow, oscillating kernel, correlations beyond M_m). The simple ι_{dyn} into the Markov limit is *falsified*; a richer embedding higher in the hierarchy (towards log-spiral e) is to be constructed.
3. **Forcing verdict (independent of 1/2).** A specificity run – the time-axis analogue of `a5_bridge_discrimination.py` – measures the distance of an *independent* invariant of the recursion section to U_3 against the matched nulls. Distance zero at U_3 and a broad band at the window nulls $\Rightarrow$ forced; otherwise allowed.

To be constructed (deliverables)

1. For ι_{geo}: represent HLV's ϕ-icosahedral projector explicitly as an operator on the $\mathbb{C}^3$ fiber ($\in \text{End}(\mathbb{C}^3)$ with C_3 invariance; Doc. 297, step 1).
2. For ι_{dyn}: compute the Markov-limit kernel of the recursion ($g_k \to 0$, Doc. 283) and its distance to U_3's hard-reset kernel against the matched-null family – the time-axis analogue of the specificity test.
3. Fix the forcing metric: distance-to-recursion-section, calibrated so that an exact section $\to 0$, a generic window $\to$ broad band.

Domain, codomain, and null family are fixed by this document; what remains open is only the construction of the three artifacts.

Counter-run performed (result, corrected)

A first version (v0.1) reduced the spiral-time window to a *winding-less* hard-reset kernel and found it carries no back-flow – which is near tautological, since a kernel without winding cannot carry a revival by construction, and it does not test Marcel's actual object. His object is defined by a *winding angle* θ (the pitch of the archimedean spiral in recursion time); the angle *is* the structure. The corrected version (`Dok299_Skripte/ffgft_299_iota_gegenlauf_winding_locked.py`, SHA256, seed 20260708) writes θ explicitly into the kernel, $k_\theta(s) = e^{-s/\tau}\cos(\theta s)$, $s = 0 \dots M{-}1$, and sweeps it. Object and witness stay apart (witness: BLP back-flow 5.125, Doc. 283); the nulls share the *same* envelope with incoherent phase – isolating the contribution of the angle.

Winding θ	Back-flow	Reading
0 (no winding)	0.00	Markov limit (in band)
$1/\phi \approx 0.618$	0.00	in band (allowed)
golden angle $\approx 137.5^\circ$	0.159	**above band**
phase-shuffled null (q95)	0.076	incoherent bound

Once the angle is in the kernel, the answer is no longer fixed by construction. At most windings the hard reset (finite width $\sim 1/M$) kills the revival ⇒ back-flow in the null band, allowed. At the *golden angle*, however, it survives the broadening and rises above the incoherent band (0.159 vs. q95 0.076); the first threshold is at $\theta_c \approx 1.82$ rad. The verdict is therefore **conditional rather than decided in advance**: allowed if the native carrier's actual winding θ^* lies below θ_c; forced (and Doc. 297 under tension at that point) if θ^* sits at the golden angle. θ^* is to be pinned from the HLV carrier – marked here as a candidate, not asserted. The absolute value (0.159) is small against the witness 5.125 and the test remains a finite proxy; but *relative to incoherent phase* it is a positive signal at that winding, not a null result. The earlier unconditional reading (v0.1) was an artifact of the winding-less reduction.

The geometric branch (ι_{geo}) is carried along as a discriminant test: the 15 pairwise axis angles against a *fixed* icosahedral reference (63.435°), not against axes taken from the carrier. Icosahedral → 0°, random-star median 21.7° – the fixed reference separates where a self-referential score would pull any star to ≈ 0. This is the anti-circularity condition a forcing test in the geometric branch must meet.

Correction value K_{frak} from the closure comma

The rolled-out recursion closes only after $N = 1/(100\xi_0) = 75$ equal pieces, not after one turn. The approximation "one turn closes" neglects a small factor: 75 turns each shortened by ε give the exact closure, $75\,(1-\varepsilon) = 74$, hence $\varepsilon = 1/75 = 100\,\xi = 0.0133\dots$ (the closure comma; as an angle $2\pi/75 = 4.8^\circ$ per turn). The per-step factor of the scale recursion $\xi_{n+1} = \xi_n(1 - 100\xi_n)$ is therefore

$$K_{\text{frak}} \;=\; 1-\varepsilon \;=\; 1 - 100\,\xi \;=\; \frac{74}{75} \;=\; 0.9867,$$

the form the corpus carries throughout (Doc. 133/231/032/285).

Important: $75 = 1/(100\xi)$ and $K_{\text{frak}} = 1 - 100\xi$ are both just 100ξ; the comma *re-expresses* the already-anchored 100ξ, it does not derive it. The anchor lies in Doc. 133 (factor 100 from the RG running, K_{frak} fixed by m_e/m_μ consistency) and in ξ itself (Higgs/α/Koide). Detailed in Doc. 300.

For the projection side (Marcel) a consistency condition follows: *if* his closed projection is a section of the same recursion, the conversion to the rolled-out curve carries the independently anchored factor $K_{\text{frak}} = 1 - 100\xi = 74/75$ – not from the 75 and not freely chosen.

Status

Working document and pre-registered forcing protocol (P35). ι remains a conjecture (Doc. 297). This document fixes the test architecture before the joint run, so that the verdict – forced or allowed – cannot be shifted after the fact. The discipline holds symmetrically: it binds even when the outcome favours FFGFT.